

Question	Answer	Marks
3(a)	(for a system in equilibrium,) sum of clockwise moments (about a point) equals sum of anticlockwise moments (about the same point)	B1
3(b)(i)	2.6×0.40 or $F \times 0.40$ (any one moment)	C1
	$U = 4.0 - F$ $= 4.0 - (2.6 \times 0.40 / 0.40) = 1.4 \text{ N}$	A1
	or	
	2.6×0.4 or $(4.0 - U) \times 0.4$	(C1)
	$2.6 \times 0.4 = (4.0 - U) \times 0.4$ hence $U = (4.0 - 2.6) = 1.4 \text{ N}$	(A1)
3(b)(ii)	$U = \rho g V$ and $A = V / h$ or $p = h \rho g$ and $A = U / p$ or $A = U / h \rho g$	C1
	$A = 1.4 / (0.10 \times 1.0 \times 10^3 \times 9.81)$	C1
	$= 1.4 \times 10^{-3} \text{ m}^2$	A1
3(c)	line starting at (0.10, 0.40)	B1
	straight line with negative gradient	B1
	line ending at (0.29, 0)	B1

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